

1. Next Greater Element:

Given an array 10, 5, 11, 6, 20, 12 use a stack to find the next greater element for each element.

2. Stock Span Problem:

Given daily stock prices 100, 80, 60, 70, 60, 75, 85 calculate the stock span day using stack.

3. From Non-Repeating character:

Given a stream of characters:

a a b c c d

Find the first non-repeating character after each insertion using a queue.

\$

```
1. #include <stdio.h>
```

```
#define MAX 100
```

```
int main()
```

```
{
```

```
    int arr[MAX], result[MAX], stack[MAX];
```

```
    int n, i, top = -1;
```

```
    printf("Enter number of elements: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter elements: ");
```

```
    for(i=0; i<n; i++)
```

```
        scanf("%d", &arr[i]);
```

```
for (int i = n-1; i >= 0; i--)
```

```
{
```

```
    while (top != -1 && stack[top] <= arr[i])
```

```
        top--;
```

```
    if (top == -1)
```

```
        result[i] = -1;
```

```
    else
```

```
        result[i] = stack[top];
```

```
    stack[++top] = arr[i];
```

```
}
```

```
printf("Next Elements Greater elements: \n");
```

```
for (int i = 0; i < n; i++)
```

```
    printf("%d → %d \n", arr[i], result[i]);
```

```
return 0;
```

```
}
```

```

2. #include <stdio.h>
   #define MAX 100
   void calculateSpan(int price[], int n, int span[]) {
       int stack[MAX];
       int top = -1;
       span[0] = 1;
       stack[++top] = 0;
       for (int i = 1; i < n; i++) {
           while (top >= 0 && price[stack[top]] <= price[i]) {
               top--;
           }
           if (top == -1)
               span[i] = i + 1;
           else
               span[i] = i - stack[top];
           stack[++top] = i;
       }
   }

   int main() {
       int price[] = {100, 80, 60, 70, 60, 75, 85};
       int n = 7;
       int span[MAX];
       calculateSpan(price, n, span);
       printf("Stock Spans: ");
       for (int i = 0; i < n; i++) {
           printf("%i ", span[i]);
       }
       return 0;
   }

```

3. #include <stdio.h>

#define MAX 100

int main () {

char str[MAX], queue[MAX];

int freq[256] = {0};

int front = 0, rear = -1;

printf ("Enter characters: ");

scanf ("%s", &str);

for (int i = 0; str[i] != '\0', i++) {

char ch = str[i];

freq[(unsigned char)ch]++;

queue[++rear] = ch;

while (front <= rear && freq[(unsigned char)queue[front]] > 1) {

```
front++;
```

```
}
```

```
if (front <= rear)
```

```
printf("After inserting %c: %c\n", ch,  
queue[front]);
```

```
else
```

```
printf("After inserting %c: -1\n", ch);
```

```
}
```

```
return 0;
```

```
}
```